Task 20: Implement branching strategies in Git by creating feature branches. Perform merging and rebasing operations, and intentionally create merge conflicts to understand and resolve them effectively

Aim

To implement **branching strategies in Git** by creating feature branches, performing **merge and rebase operations**, and resolving **merge conflicts** effectively.

Algorithm

1. Initialize Git repository.
2. Create a **new branch** (feature branch).
3. Switch to the branch.
4. Make changes and commit.
5. Switch back to main branch.
6. Perform:
    - **Merge** operation
    - **Rebase** operation
7. Create conflicting changes in two branches.
8. Attempt merge → conflict occurs.
9. Resolve conflicts manually.
10. Commit resolved changes.
11. Verify final history.

Commands Implementation

1. Create Branch

```
git branch feature1
git checkout feature1
```

2. Make Changes & Commit

```
git add .
git commit -m "Feature1 changes"
```

3. Switch to Main Branch

git checkout main

4. Merge Branch

git merge feature1

5.Rebase Example

git checkout feature1
git rebase main

6. Create Conflict

- Modify same file in **main** and **feature1**
- Then run:

git merge feature1

7. Resolve Conflict

Edit file manually:

```
System.out.println("Main branch code");

System.out.println("Feature branch code");

System.out.println("Merged code");
```

8. Final Commit

git add .
git commit -m "Resolved merge conflict"

Output:

```
MINGW64:/d/MyGitRepos/GitDemo

Deepak Panwar@DESKTOP-HV7J4HS MINGW64 /d/MyGitRepos/GitDemo (main)
$ git switch feature-1
Switched to branch 'feature-1'

Deepak Panwar@DESKTOP-HV7J4HS MINGW64 /d/MyGitRepos/GitDemo (feature-1)
$ git add .

Deepak Panwar@DESKTOP-HV7J4HS MINGW64 /d/MyGitRepos/GitDemo (feature-1)
$ git commit -m "Implemented feature-1 login"
[feature-1 820ebef] Implemented feature-1 login
 1 file changed, 1 insertion(+)
 create mode 100644 login-feature.txt

Deepak Panwar@DESKTOP-HV7J4HS MINGW64 /d/MyGitRepos/GitDemo (main)
$
```

```
Divya1@Divya:myProj [feature1] $git log --graph --oneline
* 006b9ce (HEAD -> feature1, origin/feature1) DevOps structure define
* b3378b2 small edit in lib welcome.sh
* f405de0 modified the last edit date
* 578de10 Editing the common lib file
* 1c4c923 Topic branch: iss54 created as admin user
* 6e6e9de edited temp file as an admin user
*   c366419 Merge remote-tracking branch 'origin/master'
|\
| * 4920adc edited temp file from admin user
* | feeb30c (tag: database_r1.0) modified db.log file as greets user
|/
* 9975f27 edited temp file as greets user
* 0733a3e (tag: v1.2) another new development from new-feature branch
* 55d0082 hotfix page modified from hotfix branch
*   1d67b50 Merge branch 'master' into new-feautre
|\
| * abb694b Hotfix added from hotfix branch
* | f2ff2e4 (tag: v1.1) newFeaute added from new-feautre branch
|/
*   078f9f5 Merged iss53 into master after resolving conflict
|\
| * ab3a5e5 Modified editInfo from iss53 branch
| * b777b16 modified editInfo from iss53 branch
* | 9a2412e Hotfix feature added from the hotfix branch
|/
* e69eee0 (origin/initialRemote) Cleaned up the project data files
* 86792c6 (tag: v1.0) Add the initial project structure and data
```

Result

Branches created and managed successfully Merge and rebase operations performed Merge conflicts identified and resolved Version control workflow understood